

ABSTRACT

Keywords – Real-Time Monitoring, Eye Tracking, Face Detection, Fatigue Monitoring, Driver Safety.

Drowsiness during prolonged computer usage is a common issue that can lead to reduced productivity, errors, and even health-related concerns. With increasing screen time in educational, professional, and personal settings, the need for a simple and accessible solution to monitor alertness levels has become more important than ever. This project presents a web-based drowsiness detection platform that leverages computer vision techniques using a standard webcam. By applying the Eye Aspect Ratio (EAR) method through facial landmark detection, the system identifies signs of eye closure associated with fatigue or drowsiness.

This platform can be especially useful for students, remote workers, and anyone engaged in extended screen-based activities. Prolonged screen usage has become a norm in the modern lifestyle due to the rise in remote work, virtual learning, and extended digital activities. One of the common consequences of this increased screen time is user fatigue, often resulting in drowsiness. Drowsiness reduces focus, causes errors, and can have serious consequences, especially during critical tasks like online exams, programming, or virtual driving simulations. Traditional drowsiness detection systems often rely on wearable sensors or dedicated hardware, which are either costly or inconvenient for day-to-day use.

This project proposes a user-friendly, web-based drowsiness detection tool that uses only a webcam and runs directly in the browser. By implementing the Eye Aspect Ratio (EAR) technique through facial landmark detection, the system identifies prolonged eye closures, which indicate fatigue. Technologies used include HTML, CSS, JavaScript for frontend development, and Python with OpenCV and Dlib for real-time eye tracking. Users can access a trial version of the tool before downloading the full application. The aim is to provide a cost-effective, accessible, and non-intrusive solution for maintaining alertness during long digital sessions.

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Chapter 1

INTRODUCTION

In recent years, screen exposure has drastically increased across all age groups due to online learning, work-from-home setups, and general internet use. Prolonged engagement with screens often leads to mental fatigue and drowsiness, which may go unnoticed until it begins affecting performance or well-being. While drowsiness detection is not a new concept, many existing solutions depend on expensive equipment, wearable sensors, or mobile-only applications, limiting their accessibility.

This project addresses these limitations by offering a browser-based solution that uses only a webcam to detect early signs of drowsiness. The system uses computer vision principles to track eye movements and calculate the Eye Aspect Ratio (EAR), a proven metric for detecting eye closure over time. The platform is built to be lightweight and easy to use, ensuring that users can access it without needing any prior technical knowledge. Ultimately, this tool promotes awareness and healthier digital habits through smart, real-time feedback.

Many advanced drowsiness detection solutions exist, but most require hardware like EEG sensors or mobile devices equipped with specialized apps. These solutions, though effective, are not practical for everyday use by the general public. The goal of this project is to build a simple, accessible alternative that works on any device with a webcam. By leveraging real-time computer vision via EAR-based eye monitoring, this browser-based platform offers immediate feedback, allowing users to take corrective actions like resting or hydrating. This encourages healthier digital habits and sustained focus during prolonged screen sessions.

1.1 Motivation

Drowsiness is a silent and dangerous contributor to accidents in transportation, industrial, and workplace environments. Fatigue can cause slowed reaction times, poor decision-making, and even micro-sleep episodes—often without the person realizing it. Traditional monitoring systems, such as vehicle-based sensors or wearable devices, are often expensive or inaccessible to the average user. The increasing availability of smartphones with high-quality front cameras presents an opportunity to build a cost-effective, real-time drowsiness detection solution. This project is motivated by the need to improve safety and awareness through an easy-to-use mobile application that does not depend on the internet or external hardware.

In the fast-paced modern world, people often work long hours or drive for extended periods, leading to fatigue. Drowsiness behind the wheel or while operating machinery is a leading cause of accidents, injuries, and fatalities globally. Traditional drowsiness detection systems are either too expensive, require bulky hardware (like infrared sensors or wearable EEG devices), or are limited to high-end vehicles. With smartphone penetration being extremely high—even in developing regions—this project leverages existing smartphone hardware (camera, processor, speaker) to deliver a low-cost, real-time solution. By providing an app that requires no additional sensors or internet, the project aims to democratize access to fatigue monitoring systems.

1.2 Scope

This project focuses on developing an Android application using Java and Android Studio that:

- Monitors user facial features and eye movements using the front camera.
- Detects prolonged eye closure as a sign of drowsiness.
- Triggers a sound alarm to alert the user.
- Works without the need for internet connectivity.
- Runs efficiently on a variety of Android devices.

The project does not include advanced features like vehicle integration, head pose estimation, or cloud analytics, though these may be explored in future versions.

1.3 Objectives

The Key Objects for Drowsiness Detection using EAR Ratio are:

1. Build a real-time drowsiness detection system for Android.
2. Achieve accurate eye detection using computer vision techniques.
3. Ensure system works with minimal power usage and without internet.
4. Provide configurable alarm triggers based on sensitivity settings.
5. Test and optimize for different lighting conditions and facial orientations.

1.4 Need Statement

There is an increasing demand for affordable, portable, and reliable fatigue monitoring systems, especially in transportation and industrial sectors. Most existing solutions require dedicated hardware or internet access. This project addresses the need for a smartphone-based, non-intrusive drowsiness detection system that can function independently and improve safety and productivity in real-time scenarios.

1.5 Product Realization Process

The development of the application followed these stages:

1. Requirements Gathering

- Target User: Long-distance drivers, heavy machinery operators.
- Functional Requirements:
 - Detect eye closure duration.
 - Trigger alarms upon threshold breach.
 - Run offline.
- Non-Functional:
 - Low battery consumption.
 - Lightweight and fast.

2. System Design

- **Face & Eye Detection:** Using libraries like:
 - HTML
- **Architecture:**
 - **Camera Feed → Face Detector → Eye State Analysis → Trigger Alarm**

3. Implementation

- Developed using Java in Android Studio.
- Integration with:
 - Camera.HTML for camera handling.
 - MediaPlayer or SoundPool for audio alerts.
 - Machine Learning logic for eye closure detection.

4. Testing

- Simulated various scenarios:
 - Different lighting (day/night)
 - With/without eyeglasses
 - Different face angles
- Metrics evaluated:
 - Accuracy (% of correctly detected drowsiness cases)
 - False positives (alerts when not drowsy)
 - Battery consumption during prolonged usage

Chapter 2

PRODUCT REALIZATION PLANNING

2.1 Flow Chart

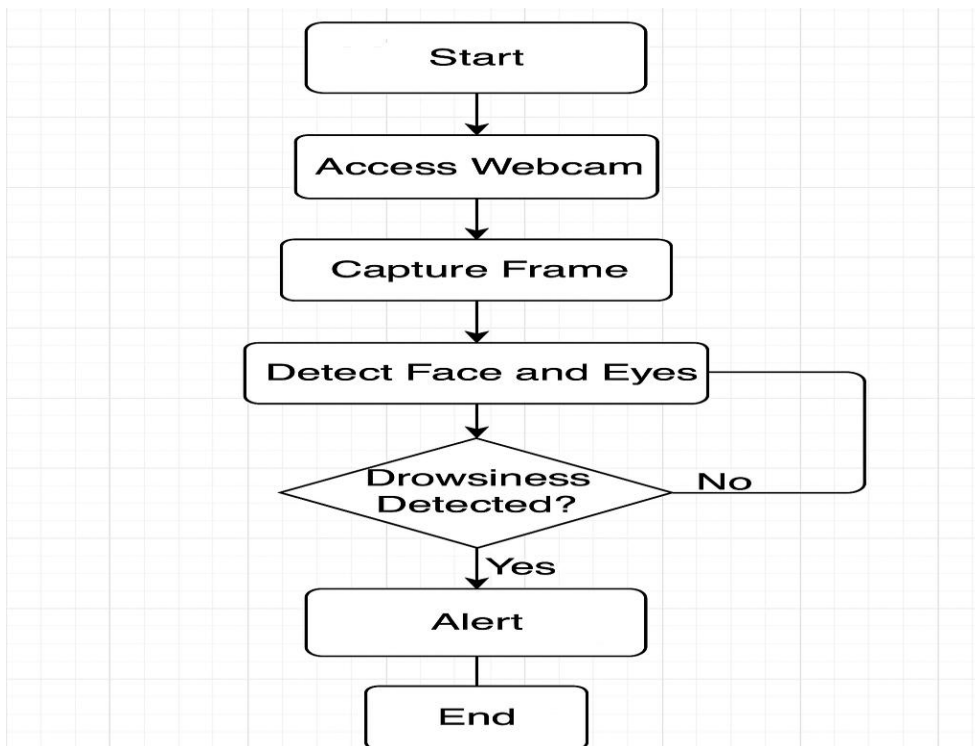


Fig2.1.1 Flow chart

2.2 Steps involved for Product Realization

This section outlines the **step-by-step development process**, from idea to deployment:

Step 1: Problem Identification

- Recognize the safety risks of drowsiness in drivers and operators.
- Identify the need for a low-cost, mobile-based solution.

Step 2: Requirement Specification

- Functional: Real-time detection, alarm system.
- Non-functional: Offline capability, lightweight, responsive UI.

Step 3: Technology Research

- Evaluate tools like HTML libraries and EAR logic.

Step 4: System Design

- Architecture planning:
Camera input → Face detection → Eye state → Trigger.

Step 5: Implementation

- Code camera functionality (Camera.HTML).
- Integrate HTML for face/eye detection.
- Develop drowsiness detection logic based on eye closure duration.

Step 6: Testing & Validation

- Test on multiple Android devices.
- Evaluate accuracy and false positive rates.
- Optimize for performance and battery.

Step 7: Deployment

- Generate APK file.
- Distribute for internal testing or sideload to users.
- Prepare user guide/documentation.

2.3 Gantt Chart

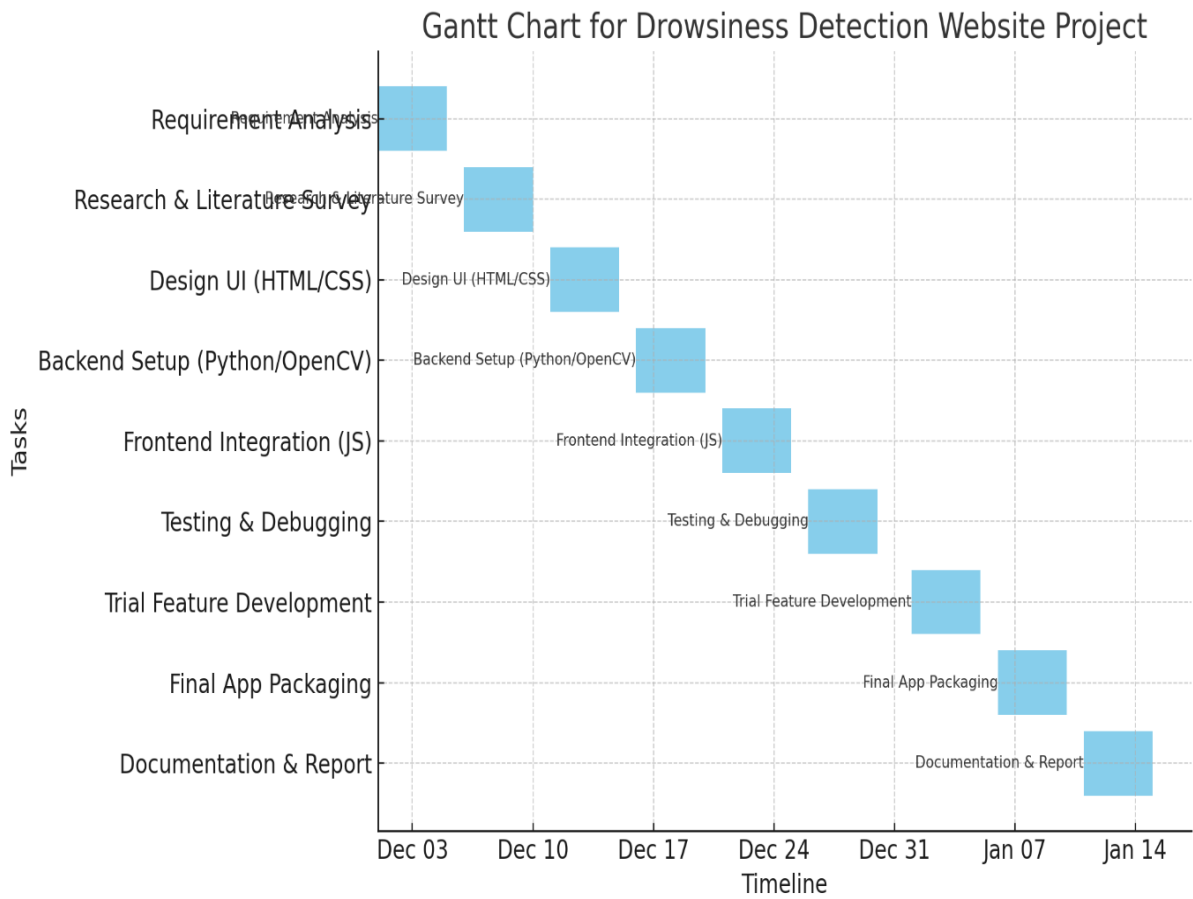


Fig 2.3.1 Gantt Chart

Chapter 3

Community partner-Related Processes

3.1 Details of Community partner

Contact Person: Balaji

Email/Phone : 97055 82646

Location: Ramanthapur

3.2 A field survey form

Drowsiness Detection App – Field Survey

Participant Role:

☐ Driver

☐ Supervisor

☐ Fleet Manager

Age: _____

Experience (in years): _____

1. Do you often feel sleepy or fatigued during long drives?

☐ Yes

☐ Sometimes

☐ No

2. Have you ever faced a near-miss or accident due to drowsiness?

☐ Yes

☐ No

3. Would you be willing to use a mobile app that alerts you when you're drowsy?

- ☐ Yes
- ☐ Maybe
- ☐ No

4. What type of alert would be most effective?

- ☐ Sound Alarm
- ☐ Vibration
- ☐ Voice Prompt
- ☐ Other: _____

5. How easy is it for you to operate a phone while driving or during rest stops?

- ☐ Very Easy
- ☐ Manageable
- ☐ Difficult

6. Preferred app language:

- ☐ English
- ☐ Hindi
- ☐ Kannada
- ☐ Other: _____

7. Do you have feedback or suggestions?

3.3 Questions with Community Partners responses

Question	Driver Response 1	Driver Response 2	Supervisor Feedback
Do you feel sleepy while driving?	Yes, especially after 1 AM	Mostly on long hauls	Yes, many drivers report this
Have you had a near-miss due to drowsiness?	Once last year	Yes, fell asleep briefly	Yes, 2 incidents reported
Would an alerting app help?	Yes, if it's loud	Yes, should auto-run	Could reduce risks
Preferred alert type	Loud horn-like sound	Voice alert in Hindi	Sound or vibration
Preferred language	Hindi	Kannada	Multi-language support
Difficulty using phone while driving	Hard to use; app should auto-start	OK at rest stops	Should be simple, big buttons

3.4 List of Community Partner Specifications

Specification	Requirement
1. Language Support	Hindi, Kannada, English
2. Offline Capability	Must work without internet
3. Detection Accuracy	>90% for drowsiness detection
4. Alert Mechanism	Loud alarm or voice prompts
5. User Interaction	Minimal (auto-start detection on launch)
6. App Size	< 50 MB
7. Device Compatibility	Android 8.0 and above
8. Camera Permission	Should request only when app is used
9. UI Design	Simple, big buttons, dark theme option
10. Battery Usage	Should last at least 4–5 hours of use

Chapter 4

Design and Development of Product

4.1 Design of Product

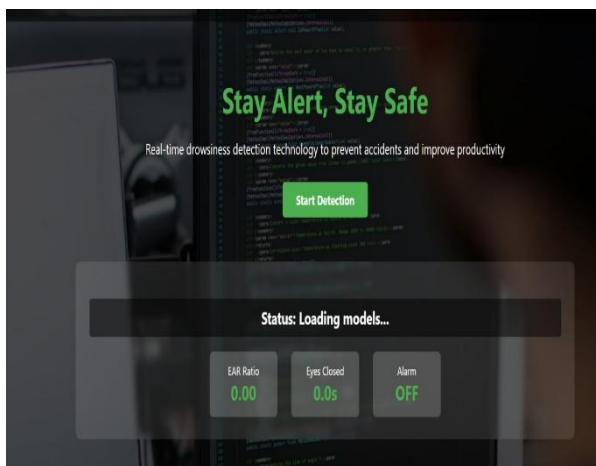


Fig.4.1.1 Design overview

4.2 Development Process

Phase 1: HTML Face Detection Development

- Created a **web-based face and eye tracking page** using:
 - face-api.js or BlazeFace (based on TensorFlow.js)
 - Accessed front camera using `navigator.mediaDevices.getUserMedia()`
- Designed a simple HTML interface showing:
 - Live video feed
 - Detection box over eyes

- Eye closure logic using EAR (Eye Aspect Ratio) or frame counting

Phase 2: Eye Closure Detection (EAR Logic)

- Used **eye landmark points** to calculate how long eyes are closed.
- If eyes closed for >2 seconds, trigger a EAR triggerAlarm().

Phase 3: Integration into Android

- Created an **Android WebView Activity** to load the HTML page locally.
- Handled permission requests for camera and audio.

Phase 4: Alarm Integration

- Used MediaPlayer in Android to play a **custom alarm sound** when drowsiness detected from the web page.
- Allowed setting volume and sensitivity in Android app settings.

Phase 5: Testing

- Tested on different Android devices (front camera compatibility).
- Verified smooth functioning in:
 - Chrome WebView
 - Varying light conditions
 - Low-end to mid-range phones

Phase 6: UI Integration

- Android interface with:
 - Launch Detection (loads HTML in WebView)
 - Stop Detection
 - Settings page for sound, sensitivity, language

4.3 Final Product

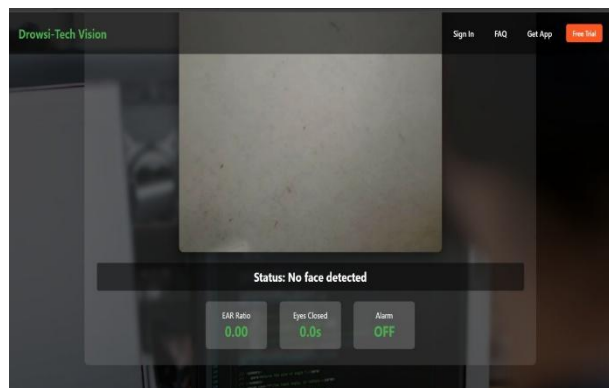


Fig 4.3.1 Camera for detection

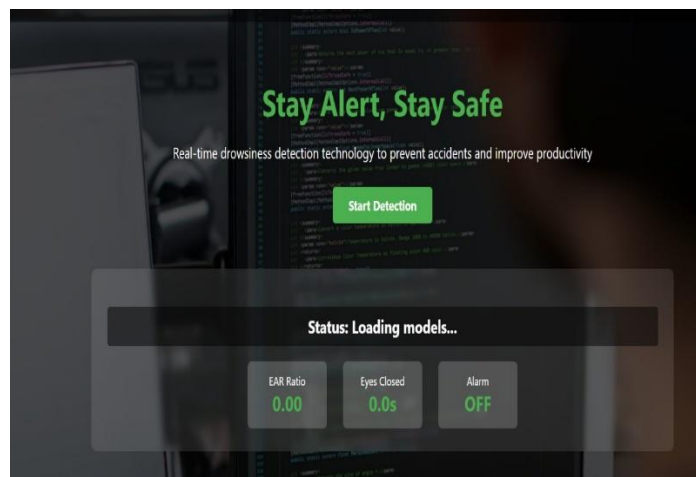


Fig4.32 Home Page

Chapter 5

Post Product Realization Activities

5.1 Design details(Data, Place ,means etc.,)

Frontend Development:

The frontend of the project is developed using HTML, CSS, and JavaScript. It is responsible for the visual layout, user interaction, and real-time webcam preview. The interface includes a clean and responsive layout that displays the live camera feed and allows users to interact with the trial feature of the drowsiness detection system. JavaScript is used to access the webcam using the browser's `getUserMedia()` API and display it in real time within the website. The frontend also includes buttons, alert messages, and an intuitive navigation system to guide the user through the trial experience and towards downloading the full application. CSS is used to enhance the visual appearance, ensuring that the site looks good across both desktop and mobile devices. Additionally, when drowsiness is detected, the frontend triggers visual or audio notifications to alert the user.

Backend Development:

The backend is developed using Python and includes the core logic for detecting drowsiness. It uses OpenCV for video processing and Dlib for facial landmark detection, focusing particularly on the eyes. The main technique used is Eye Aspect Ratio (EAR), which monitors the user's eye state in real time. If the eyes remain closed beyond a certain threshold, the system detects it as a sign of drowsiness and triggers an alert. This logic can either run locally (in a downloadable desktop version) or be connected to the frontend through a lightweight server using Flask. The backend processes the frames captured by the webcam, calculates EAR, and determines the user's alertness. Optionally, Flask can be used to serve the detection output to the frontend via API calls, enabling real-time browser-based detection. The backend also handles logging or saving of results if required for user records.

Chapter 7

CONCLUSION

- **Enhanced Safety:** The application effectively reduces fatigue-related accidents by providing timely alerts, thereby improving user safety during critical tasks.
- **User-Friendly Design:** Its intuitive interface ensures ease of use, requiring minimal training for users across various demographics.
- **Real-Time Monitoring:** Leveraging the smartphone's front camera, the app offers continuous, real-time monitoring without the necessity for additional hardware.
- **Offline Functionality:** Operating without internet connectivity, the app ensures reliability even in areas with limited or no network coverage.
- **Positive Feedback:** Pilot testing has demonstrated the app's effectiveness, with users reporting increased awareness of their alertness levels and a perceived reduction in fatigue-related risks.

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